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Course: Python 100 Assignment 06_ Model 06

Assignment 06 - Python Programming Basics

Module 06 - Collections of data

Introduction

Create a Python program that demonstrates using constants, variables, and print statements to display a message about a student's registration for a Python course. This program is very similar to Assignment05, but **It adds the use of functions, classes, and using the separation of concerns pattern.**

File Name:

- The file is named Assignment06.py

Script Header:

- The script header includes this text and has been updated with your name and the current date.

Constants:

- **The constant MENU: str is set to the value:**

```
---- Course Registration Program ----
```

```
Select from the following menu:
```

```
1. Register a Student for a Course
```

```
2. Show current data
```

```
3. Save data to a file
```

```
4. Exit the program
```

```
-----
```

- **The constant FILE_NAME: str is set to the value "Enrollments.csv"**
- **The constant values do not change throughout the program.**

Variables:

- **menu_choice: str is set to empty string.**
- **students: list: list is set to and empty list**

Classes:

- The program includes a class named **FileProcessor**.
- The program includes a class named **IO**.
- All classes include descriptive document strings.

Functions:

- All functions include descriptive document strings.
- All functions with except blocks include calls to the function handling error messages.
- All functions use the **@staticmethod** decorator.
- The program includes functions with the following names and parameters:
- **output_error_messages(message: str, error: Exception = None)**
- **output_menu(menu: str)**
- **input_menu_choice()**
- **output_student_courses(student_data: list)**
- **input_student_data(student_data: list)**
- **read_data_from_file(file_name: str, student_data: list):**
- **write_data_to_file(file_name: str, student_data: list):**

Input / Output:

- On menu choice 1, the program prompts the user to enter the student's first name and last name, followed by the course name, using the **input()** function and stores the inputs in the respective variables.
- Data collected for menu choice 1 is added to the students two-dimensional list of dictionaries rows.
- On menu choice 2, the program uses the **print()** function to show a string of comma-separated values for each row collected in the students variable.

Summary: Please refer to the screenshots

1, run the script to display the menu; choose 2 to show the error handling for empty Jason file

```

C:\Users\frank\PyCharmMiscProject\.venv\Scripts\python.exe "C:\Users\frank\PyCharmMiscProject\homework result\Assignment06_F

---- Course Registration Program ----
Select from the following menu:
1. Register a Student for a Course
2. Show current data
3. Save data to a file
4. Exit the program
-----

Enter your choice: 2
-----
No registrations found.
-----

```

2, choose 1 and input blank values to show the error handling for blanks

```

Enter your choice: 1
Enter student's first name:

Error:
There was a problem entering data.
-- Technical Error Message --
Inappropriate argument value (of correct type).
First name should contain letters only.

```

```

Enter your choice: 1
Enter student's first name: Frank
Enter student's last name:

Error:
There was a problem entering data.
-- Technical Error Message --
Inappropriate argument value (of correct type).
Last name should contain letters only.

```

3, choose 1 and input correct values; and then choose 2 to show the inputs

```
-----  
Enter your choice: 1  
Enter student's first name: Frank  
Enter student's last name: Li  
Enter course name: Python 003  
  
Registered Frank Li for Python 003
```

```
---- Course Registration Program ----  
Select from the following menu:  
1. Register a Student for a Course  
2. Show current data  
3. Save data to a file  
4. Exit the program  
-----
```

```
Enter your choice: 2  
-----
```

```
Frank, Li, Python 003  
-----
```

4, choose 3 to save the inputs into Jason file

```
Enter your choice: 3
```

```
Data successfully saved.
```

```
-----  
Frank, Li, Python 003  
-----
```

```
File Edit View
[
  {
    "FirstName": "Frank",
    "LastName": "Li",
    "CourseName": "Python 003"
  }
]
```

5, choose 1 to add more input; and choose 3 to save the inputs

```
Enter your choice: 1
Enter student's first name: Frank
Enter student's last name: Li
Enter course name: C# 003

Registered Frank Li for C# 003
```

```
-----
Enter your choice: 2
-----
Frank, Li, Python 003
Frank, Li, C# 003
-----
```

```
Enter your choice: 3

Data successfully saved.

-----
Frank, Li, Python 003
Frank, Li, C# 003
-----
```

```
File Edit View

[
  {
    "FirstName": "Frank",
    "LastName": "Li",
    "CourseName": "Python 003"
  },
  {
    "FirstName": "Frank",
    "LastName": "Li",
    "CourseName": "C# 003"
  }
]
```

6, choose 4 to quit the menu

```
Enter your menu choice: 4

Thank you for using the Course Registration Program.
Program ended.

Process finished with exit code 0
```

Result:

- The program takes the user's input for a student's first, last name, and course name.
- The program displays the user's input for a student's first, last name, and course name.
- The program saves the user's input for a student's first, last name, and course name to a JSON file. (check this in PyCharm or with a simple text editor like Notepad or TextEdit.)
- The program allows users to enter multiple registrations (first name, last name, course name).
- The program allows users to display multiple registrations (first name, last name, course name).
- The program allows users to save multiple registrations to a file (first name, last name, course name).
- The program runs correctly in both **PyCharm** and from the **console or terminal**.

The script can handle the errors below:

- The program provides structured error handling when the file is read into the list of dictionary rows.
- The program provides structured error handling when the user enters a first name.
- The program provides structured error handling when the user enters a last name.
- The program provides structured error handling when the dictionary rows are written to the file.